

Daily Logic Drill #5

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Spotting Flawed Steps · Common Fallacies (affirming the consequent, denying the antecedent, unjustified division) · Line-by-Line Proof Critique.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Name the fallacy, or say whether the reasoning pattern is valid. Pure recognition.

1. “ $P \Rightarrow Q$, and Q is true; conclude P is true.” Is this reasoning valid?
2. “ $P \Rightarrow Q$, and Q is false; conclude P is false.” Is this reasoning valid?
3. “ $P \Rightarrow Q$, and P is false; conclude Q is false.” Is this reasoning valid?
4. “ $P \Rightarrow Q$, and P is true; conclude Q is true.” Is this reasoning valid?
5. Spot the flaw: “ n is odd $\Rightarrow n^2$ is odd. $n^2 = 49$ is odd. Therefore $n = 7$.”
6. Spot the flaw: “If a shape is a square, it’s a rectangle. This shape is not a square. Therefore it’s not a rectangle.”
7. True or false: “If $P \Rightarrow Q$ and $Q \Rightarrow P$ are both true, then assuming Q to conclude P becomes valid reasoning.”
8. Divide both sides of $x^2 = 5x$ by x to get $x = 5$. Which solution got lost?
9. True or false: “Squaring both sides of an equation can introduce extraneous (fake) solutions.”
10. True or false: “Dividing both sides of an equation by a variable expression can lose valid solutions.”

Section B Manipulation Drills

(≤50 s each)

Find the specific broken step in a short argument — or confirm there isn’t one.

1. Spot the flaw: “ n^2 is a multiple of 4. Therefore n is a multiple of 4.”
2. Spot the flaw: “ $\sqrt{x^2} = x$ for all real x .”

3. A “proof” that $a = 0$ for any $a = b$: Line I: $a = b$. Line II: $a^2 = ab$. Line III: $a^2 - b^2 = ab - b^2$. Line IV: $(a - b)(a + b) = b(a - b)$. Line V: dividing both sides by $(a - b)$: $a + b = b$, so $a = 0$. Which line introduces the flaw, and why?
4. Spot the flaw: “If it is sunny, I wear sunglasses. I am wearing sunglasses. Therefore it is sunny.”
5. Spot the flaw: “ $x^2 > 4$. Therefore $x > 2$.”
6. Spot the flaw: “If n is a multiple of 6, then n is a multiple of 2 and of 3. n is a multiple of 2. Therefore n is a multiple of 6.”
7. Is there a flaw in this argument? “ n is odd (given). Line 1: $n = 2k + 1$ for some integer k . Line 2: $n^2 = 4k^2 + 4k + 1$. Line 3: $n^2 = 2(2k^2 + 2k) + 1$. Line 4: therefore n^2 is odd.”
8. Spot the flaw: “All multiples of 4 are even. n is even. Therefore n is a multiple of 4.”
9. Spot the flaw: “ $\sqrt{4} = \pm 2$, so $\sqrt{4} = 2$ and $\sqrt{4} = -2$ simultaneously, meaning $2 = -2$.”
10. Spot the flaw: “Since $(-2)^2 = 2^2 = 4$, we can cancel the squares to conclude $-2 = 2$.”

Section C Substitution & Structure

(≤90 s each)

Longer arguments — some flawed, some genuinely correct. Decide which, and why.

1. Consider this argument. Line I: let n be even, so $n = 2k$. Line II: $n^3 = 8k^3$. Line III: $n^3 = 2(4k^3)$. Line IV: therefore n^3 is even and n^3 is a multiple of 8. Which best describes this attempt?
 - A) Completely correct.
 - B) Error in line II.
 - C) Error in line III.
 - D) The conclusion “ n^3 is a multiple of 8” does not follow.
2. “Claim: if n is a positive integer and $n^2 > 100$, then $n > 10$. ‘Proof’: $n^2 > 100 \Rightarrow n > 10$ (taking square roots of both sides).” Is this proof valid? If not, give a counterexample.
3. “Claim: if a, b are integers with $a \mid b$ and $b \mid a$, then $a = b$.” Spot the flaw and give a counterexample.
4. “Claim: for positive real x , if $x + \frac{1}{x} = 2$, then $x = 1$. ‘Proof’: $x + \frac{1}{x} = 2 \Rightarrow x^2 + 1 = 2x \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x - 1)^2 = 0 \Rightarrow x = 1$.” Is this proof valid?
5. “Claim: if n is an integer and n^2 is divisible by 4, then n is divisible by 4. ‘Proof’: n^2 divisible by 4 $\Rightarrow n^2 = 4m$ for some integer $m \Rightarrow n = 2\sqrt{m}$, and since n is an integer this is always even, hence n is divisible by 4.” Identify exactly where this breaks down, and give a counterexample.
6. “Claim: the equation $x^3 = x$ has exactly one real solution. ‘Proof’: $x^3 = x \Rightarrow x^3 - x = 0 \Rightarrow x(x^2 - 1) = 0 \Rightarrow x = 0$ (dividing by $(x^2 - 1)$).” Spot the flaw.

7. “Claim: if p is prime and $p > 2$, then $p + 1$ is not prime. ‘Proof’: p is odd (Day 1, B9), so $p + 1$ is even. Since $p + 1 > 2$ is even, it cannot be prime, as the only even prime is 2.” Is this proof valid?
8. “Claim: for every positive integer n , $n^2 \geq 2n - 1$, with equality only at $n = 1$. ‘Proof’: $n^2 - 2n + 1 = (n - 1)^2 \geq 0$ always, with equality exactly when $n = 1$.” Is this proof valid?

Section D Challenge

(TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

- A student is asked to prove: “If n is even, then n^2 is even.” The student instead writes: “Suppose n^2 is even. Then $n^2 = 2m$ for some integer m . If n were odd, n^2 would be odd (Day 1, B5) — contradiction. So n is even.” Which best describes this attempt? (*after TMUA 2021 Paper 2 Q11 format*)
 - A completely correct proof of the original statement.
 - Incorrect — the student has proved the converse of the original statement instead.
 - Incorrect because there is an error in the contradiction step.
 - Incorrect because odd numbers were not properly defined.
- A student “proves” that $\sqrt{2}$ is rational as follows: “Suppose $\sqrt{2} = p/q$ in lowest terms. Then $2 = p^2/q^2$, so $2q^2 = p^2$, so p^2 is even, so p is even, so $p = 2r$. Then $2q^2 = 4r^2$, so $q^2 = 2r^2$, so q is even too. But p, q were in lowest terms, so they can’t both be even — contradiction. Hence $\sqrt{2}$ is rational.” Spot the flaw in the final line’s conclusion.
- “Reasoning: for every positive integer n , $n^2 + n$ is even; therefore, by strong induction, every polynomial in n with integer coefficients is even.” What is wrong with this reasoning, independent of whether the individual claim “ $n^2 + n$ is even” is itself true?
- “Claim: a triangle has two equal angles if and only if it has two equal sides (isosceles); therefore, having two equal angles is logically equivalent to having a right angle.” What is wrong with this argument?
- A student presents this argument: “Claim: if n is a positive integer with $n^2 - n + 41$ composite, then $n \geq 40$. Proof: we showed (Day 4, C1) that $n^2 + n + 41$ is prime for $n = 0, \dots, 39$ and composite at $n = 40$. Since $n^2 - n + 41$ is just $n^2 + n + 41$ with n replaced by $-n$, the same primality pattern applies, so $n^2 - n + 41$ is composite exactly when $n \geq 40$ too.” Identify precisely what is wrong with this argument, and give a specific counterexample.

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths